

Call for Expressions of Interest

<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	IBM AI Racing League: Build and Race an Autonomous AI Driver
<i>Company:</i>	IBM

Contact Details

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Additional Info

<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☐ If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Artificial intelligence; autonomous systems; simulation; Python programming; software engineering; optimisation; human-AI collaboration; game/simulation development; responsible AI; technical communication.

Project Background Information

This project is based on the IBM AI Racing League Winter mission briefing. The purpose is for student teams to design, build, test, and race an autonomous AI driver using TORCS, Python, and IBM Bob. The project combines practical software engineering, simulation, AI-assisted development, optimisation, and competitive demonstration.

The core mission is to build a high-performance AI race car using a combination of physics simulation, Python scripting, and generative AI logic. TORCS, the Open Race Car Simulator, provides the physics engine, track environment, and vehicle dynamics. Python provides the scripting and algorithm environment where the driving logic is created and executed. IBM Bob, accessed via SkillsBuild, is used as the AI development assistant to troubleshoot logic, write and improve code, and support software development.

Students will start from a baseline driver and iteratively improve its performance. The briefing identifies a cornering fault in the baseline script and positions IBM Bob as the “crew chief” that can help analyse code, identify missing braking/steering strategies, and recommend corrections. The project therefore gives students a realistic optimisation loop: branch from a safe base, discuss code and architecture with IBM Bob, implement updates, test in TORCS, and validate lap-time performance.

The competitive target is to race one lap from a standing start on the Corkscrew track. Teams should aim to produce a robust driver that can perform on track, cope with race conditions, and, as a stretch

¹ An overview of different research areas in the school of computer science can be found [here](#).

goal, handle F1-style multi-car races. The project also requires students to explain how they used IBM Bob and to submit evidence of their engineering journey.

Beyond the Corkscrew

The students should code their AI logic to deal with overtaking, and avoiding other cars. Also although the students can test their car on Corkscrew – the car should be capable of navigating any course on TORCS

Final step

The final deliverable will include a video multi car – multi player race – on Corkscrew demonstrating the cars track performance inc overtaking and car avoidance

Ultimate brownie points

Encourage a course wide competition – video the highlights and the final race with the fastest cars competing. Winners to receive branded IBM merch and a place in the European Finals

Requirements

The group must provide:

- A Python-based autonomous racing driver for TORCS, created by improving the permitted driver script.
- A functioning implementation that can complete one lap from a standing start on the Corkscrew track.
- A development approach that uses IBM Bob as the AI co-pilot throughout requirements analysis, debugging, code improvement, testing, and documentation.
- Evidence that IBM Bob was used as an AI “crew chief” to analyse driver logic, identify weaknesses, suggest improvements, and support technical decision-making.
- A record of the optimisation loop used by the team: branch, discuss with IBM Bob, implement, test in TORCS, and validate.
- A short technical explanation of the driver strategy, including steering, braking, acceleration, cornering, recovery, and any racing-line or overtaking logic.
- A test summary showing how the car performed in TORCS, including lap-time evidence where available.
- A one-lap standing-start video on the Corkscrew track.
- A team video explaining how the team used IBM Bob during the project.
- The final Python file, named using the team naming convention agreed with the project lead.
- A short blog or field report documenting the engineering journey, challenges, breakthroughs, and lessons learned.
- A GitHub repository or equivalent source-code submission containing the final code, setup notes, test evidence, and project documentation.

Rules and constraints:

- Teams may have no more than five members.
- Multiple teams from a university are permitted.
- Teams must only update the permitted Python driver file, identified in the briefing as snakeoil3_gym.py.
- Teams must use IBM Bob as the AI co-pilot.
- Submissions must be Python code only; no external models should be submitted.
- Drivers must not be intentionally coded to block following cars or ram other cars off the track.
- Drivers must not intentionally leave the track to cut corners.
- Teams should factor in potential F1-style multi-car racing, including overtaking, returning to track if nudged, and avoiding unsafe blocking behaviour.
- If using RaceRunner or multi-car configuration, teams must ensure the Python code uses the correct 300x port to match the corresponding scr_serverx car in TORCS.
- All team members must complete the Innovation Award before registering their team.

Expected outcomes:

- A working AI racing driver that can complete the required TORCS lap challenge.
- Clear evidence of iterative AI-assisted optimisation using IBM Bob.
- A race-ready technical artefact: one Python file, lap video, team video, blog/field report, and supporting documentation.
- Improved student confidence with Python, simulation, AI-assisted software development, testing, and optimisation.
- A project suitable for showcase activity across university cohorts, Discord communities, and potential multi-car race events.

Useful resources from the briefing:

- TORCS Quick Start Guide: <https://ibm.box.com/v/TORCSQuickStartExt>
- TORCS download assets: <https://ibm.box.com/v/TORCSdownloadzip>

- IBM Bob University access: <https://ibm.biz/IBMBobUniversity>
- IBM Bob Python introduction: <https://ibm.biz/introbobpython>
- IBM Bob update/support material: <https://ibm.biz/bobupdate>
- Innovation Award certificate: <https://ibm.biz/IBMInnovationCertificate>
- Team registration: <https://ibm.biz/RaceatWinter>
- Submission portal: <https://ibm.biz/wintersubmission>
- Discord community: <https://discord.gg/AJyJRbWmDw>
- RaceRunner configuration video: <https://youtu.be/NFBez8Wbzpl>
- Getting started video: <https://youtu.be/rRbpOUH0IFQ>

Skills

Highly desirable: • Python programming and debugging. • Interest in autonomous systems, simulation, racing, or optimisation. • Ability to work with TORCS or similar simulator environments. • Practical use of IBM Bob as an AI development assistant. • Git/source control, testing discipline, and clear technical documentation.	Desirable: • Experience with game/simulation development. • Understanding of control logic, racing lines, steering, braking, and acceleration strategies. • Familiarity with algorithmic optimisation or basic AI decision-making. • Video/demo creation and technical blogging. • Experience working in teams and explaining design decisions clearly.
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Submission / Award information

<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.